



SJAA EPHEMERIS

SJAA Activities Calendar

Jim Van Nuland

August

- 4 Houge Park star party. Sunset 8:12 p.m., 79% moon sets 1:44 a.m. Star party hours: 9:30 to 12:00
- 5 ATM Workshop at Houge Park. 7:30 p.m.
- 12 **General meeting at Houge Park.** Timothy Thompson of NASA/JPL on Infrared Astronomy and the Spitzer Space Telescope. 8 p.m.
- 17 ATM Workshop at Houge Park. 7:30 p.m.
- 18 Astronomy Class at Houge Park. 7:30 p.m.
- 18 Houge Park star party. Sunset 7:56 p.m., 18% moon rise 2:10 a.m. Star party hours: 9:00 to 12:00
- 19 Coyote Lake Star Party. Sunset 7:54 p.m. 11% moon rises 3:11 a.m.
- 26 Dark sky weekend. Sunset 7:45 p.m., 10% moon sets 9:06 p.m.
- 26 AANC Star-B-Q at Fremont Peak State Park

September

- 1 Houge Park star party. Sunset 7:36 p.m., 64% moon sets 0:25 a.m. Star party hours: 8:30 to 11:30
- 2 ATM Workshop at Houge Park. 7:30 p.m.
- 9 **General meeting at Houge Park.** Slide and Equipment night. Latest images, latest equipment. 8 p.m.
- 14 ATM Workshop at Houge Park. 7:30 p.m.
- 15 Astronomy Class at Houge Park. 7:30 p.m.
- 15 Houge Park star party. Sunset 7:15 p.m., 31% moon rise 1:04 a.m. Star party hours: 8:00 to 11:00
- 16 Dark sky weekend. Sunset 7:14 p.m., 23% moon rise 2:07 a.m.
- 21/24 CalStar star party at Lake San Antonio Park
- 23 Dark sky weekend. Sunset 7:03 p.m., 2% moon sets 7:33 p.m.
- 29 Houge Park star party. Sunset 6:54 p.m., 48% moon sets 11:14 p.m. Star party hours: 8:00 to 11:00
- 30 ATM Workshop at Houge Park. 7:30 p.m.
- 30 Coyote Lake Star Party. Sunset 6:52 p.m. 49% moon sets 0:19 a.m.

The Board of Directors meets at 6:00 p.m. preceding each general meeting. All are welcome.

August General Meeting

Science & Images from the Spitzer Space Telescope - Timothy Thompson

August 12, 2006 at 8 p.m.

David Smith

The infrared Spitzer Space Telescope, launched in August 2003, is the last of NASA's Great Observatories. Tim will briefly describe the telescope & its instruments, and why we want to do infrared astronomy. Then he will review the images and science that have come from the Spitzer Space Telescope in its first 3 years of operation. Timothy has been on the board of directors of the Mt. Wilson Observatory Association for about 22 years. He is President Emeritus of the Los Angeles Astronomical Society, and a veteran of 17 years on the LAAS board of directors. He has an MS degree in physics from Cal State L.A., and has been in the science division at JPL for 25 years. Currently he works in the Center for Long Wavelength Astrophysics, analyzing images from the Spitzer Space Telescope. He is developing techniques for enhancing the resolution of images from the IRAC & MIPS instruments on Spitzer. His research interests also include creating chemical and radiative transfer models of the outer planet atmospheres, and studying the Jovian plasma environment.

CalStar 2006

Rob Hawley

The 2006 edition of San Jose Astronomical Association (SJAA)'s annual star party CalStar will be held at Lake San Antonio County Park on the nights of September 21 through the 23 (Thursday through Saturday). The web site describing the event is available at <http://www.sjaa.net/calstar>. The website features a lot of new information such as a picture tour of the CalStar site, advice for first time visitors, and a new layout of the viewing areas. Be sure to check it out. Registration opens August 20.

This year we will be expanding our casual viewing area to offer more alternatives to the CalStar experience. We will once again encourage people that will not be staying overnight to use special areas set aside for early departure.

For more information email calstar@sjaa.net.

24 hour news and information hotline: (408) 559-1221

<http://www.sjaa.net>



Chabot aerial: The Center is now located in the Oakland hills.



Observatory director Charles Burkhalter (1886 -1923) looks through the 20-inch telescope "Rachel", which was installed in 1916 and still in use today.

Chabot Observatories Exhibit – Opens August 5, 2006

Judyth Collin

Since 1883, countless visitors have gazed through the Chabot telescopes at the wonders of the night sky. A new exhibit in the observatory domes explores the history of the Chabot Observatories and how its historic telescopes continue to be used today. Daytime visitors can virtually operate a telescope, experiment with mirrors and lenses to understand how telescopes create images of distant objects, and travel through more than a century of Chabot's history via multimedia kiosks, historical images, and artifact displays.



The 20-inch refractor telescope, Rachel, installed in 1915, was refurbished for its move to the new site on Skyline Blvd.



The original 1883 8-inch telescope, named "Leah", is still in use today. All pictures on page 2 and 3 of this issue are from Judyth Collin and the Chabot Observatory.



Chabot's newest telescope, a 36-inch reflector, named "Nellie", opened in June, 2003.



Three observatories house Chabot's telescopes.

Chabot and the Amateur Community

Celeste Burrows

Chabot offers a valuable free service to amateurs and to the general public, and that is year-round observing every Friday and Saturday night through Chabot's three large telescopes as well as through volunteer telescopes out on the observing deck. Installation of a research grade solar telescope is in our future and eventually likely a radio dish also.

Amateurs are the operators for Chabot's telescopes in our evening public programs; they also conduct public observing of the sun, moon, or Venus during most weekend days. Amateur volunteers also do most of the equipment maintenance.

Members of Chabot or of the Eastbay Astronomical Society have additional observing opportunities, including private viewing nights. EAS volunteers may be trained to operate the observatory telescopes for the public or for their own use on designated nights. Several amateur observing projects are going on now. The exoplanet group uses Chabot's telescopes to track HD209458 and eight transit candidates, in conjunction with Lick and UCB. An astrophotography group uses Chabot's telescopes and their own to learn and practice astrophotography. Their photos appear on EAS' website and are displayed inside the domes.

Volunteers have additional access to Chabot's programs, including training on telescopes (either the observatory's or other amateur scopes) astronomy and observing workshops and speakers, and other events at the Center. Commitment for volunteers is 8 hours per month at the tasks of their choosing.

Observing also takes place following

the monthly EAS meeting at Chabot. Currently we meet monthly on the second Saturday but may switch some meetings to Sundays and this will permit more private telescope time after the meeting. Family observing time may be offered before those meetings during standard time months.

Chabot proudly hosts the telescope makers workshop which has served the local community for more than 70 years. This workshop, offered free Friday nights



The original Chabot observatory, which opened in 1883, was located in downtown Oakland, California.

year round, supports amateurs and provides local youth with an entrée into astronomy as well as a lifelong avocation.

An active teen program also supports the development of local youth. Here teens get ongoing training in science, astronomy, telescope operation, and personal skills, and use these to work with our public at the Center and through outreach to a variety of schools and neighborhood centers during the year.

Observing continues inside with the Zeiss planetarium, the best dark sky within a day's trip. Natural dark skies are shown to the public, perhaps the first that they've seen, on Saturday nights and are used for scouts' badge workshops and our occasional introductory astronomy classes. Bring

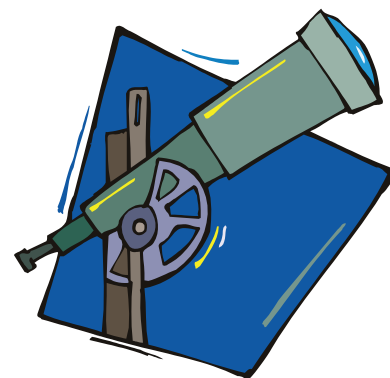
your binoculars when you come, for 18 Messier objects can be found in this indoor sky.

The new full-dome digital projection system permits trips to anywhere in the universe, on Friday nights, using real data from the current galaxy surveys and real images from Solar System missions.

Weekdays the next generation of amateur and professional astronomers is being trained on their school field trips to Chabot. Hundreds of children each day visit the telescopes and other exhibits, see space shows and take workshops using materials such as those developed by Project and Family Astro or the Night Sky Network. Teachers are trained on content and hands-on activities in summer workshops.

Chabot's 123-year commitment as a free public observatory is maintained in part through the active involvement of amateurs, in operating our telescopes and in serving on boards and committees. If there are ways we can better serve the amateur community do not hesitate to write and suggest them.

Celeste Burrows is the astronomy instructor for Chabot and specializes in teaching the student and teacher workshops. She operates her own telescopes and Chabot's in her spare time. She is also on the board of EAS.



Uranus and Neptune are Bigger Than The Moon, Too

Akkana Peck

August is a slow month for observing the planets.

Jupiter is still observable low in the southwest at dusk, and remains up for a few hours, though it's low in the sky and past its prime. It's been a great Jupiter year (even if I never did manage to see that darned Red Spot Junior) -- there's been lots of detail to watch in the bands and festoons.

There are lots of double shadow and moon transits on Jupiter this month, too. A shadow transit is when the shadow of one of Jupiter's four largest "Galilean" moons falls on the planet as we see it from Earth. They're fairly easy to see even when Jupiter is low in the sky, even with a small telescope -- I've seen them using the club's 60mm loaner refractor. The Galilean moons move fast: It only takes a few hours for one of their shadows to cross Jupiter's disk. So you can watch ingress (when the shadow first appears) to egress (when it disappears off the other side), and in many cases you'll also get to watch the moon that's casting the shadow cross in front of Jupiter, where it will eventually disappear against the planet's brightness.

But they're usually visible for a while against the limb of Jupiter before they disappear, and if you haven't seen that, it's an eerie sight every astronomer should see at least once.

But really, this month is best for the outer planets. Neptune is at opposition on August 11th, and is very well placed for observing. Look for it in Capricornus. Uranus, too (in Ophiuchus) is well placed for observing. And Pluto is just a few months past opposition, so this is still prime Pluto season: grab a good chart and a big scope and go at it.

Mercury emerges into morning twilight early in the month, but is lost again

by mid-month. Venus, too, is in the morning sky, and makes a close pass with Mercury on the 10th, and a very close pass with Saturn (which is only visible during the latter half of the month) on the 26th and 27th.

Mars is too close to the sun to be easily observed right now. That's ironic, because it looks like the annual flood of "Mars will be BIGGER THAN THE MOON!" hoax emails has begun to flow (inspired by the conjunction three years ago, but people never put dates or details on stuff they forward around the internet, do they?) I talked about this hoax in last August's column, while we were all getting ready for the Mars opposition. It didn't occur to me at the time that I'd see it again, even in years like this one that don't have a Mars opposition at all; but I guess this may become an annual event.

Oh, you were wondering about the title of this column? Well, last year when the hoax mails were flying, people kept asking me "Is Mars going to be bigger than the moon?" The answer to that is, of course Mars is bigger than the moon. And Uranus and Neptune -- well, they're WAY bigger than our little moon. In fact, all the planets except Pluto are bigger than our moon. So if someone asks you if Mars is going to be bigger than the moon this August ... the answer is yes! It just won't LOOK bigger ...



Some Notes on Sky and Telescope Subscriptions

Rob Hawley

Recently (early July) members whose subscriptions expire in October have been receiving an alarming letter from S&T indicating their subscription has expired and they may already have missed an issue.

Well it simply isn't true.

If your subscription expires in October it means you will receive your final issue at the end of August. They don't even run the mail lists for the October issue until early August based on the information I have received.

I complained to S&T last week and then I received a letter for *my* subscription. My August issue arrived the previous week and I know that two more copies are coming.

If I have your email address then you get an email from me about 90 days before the expiration date. For example, the October expiring members received an email in late June. If you reply to that email promptly we will get your renewal to S&T before they really need it. I could move the notification back another month, but that would not insure that people did not receive these notices.

So what do you do if you get one? Well, if you received this letter you already have gotten an email from me anyway so you should renew now. (This is a good reason to insure that SJAA has your current email address).

As far as the lost issues, etc. I would suggest taking a deep breath.

If anyone has any concerns they can contact me at membership@sjaa.net. If you would like to have your club membership expire in the month that you get the notice for S&T then contact me and that can be arranged. In practice it does not really matter.

From Thunderstorms to Solar Storms...

By Patrick L. Barry

When severe weather occurs, there's a world of difference for people on the ground between a storm that's overhead and one that's several kilometers away. Yet current geostationary weather satellites can be as much as 3 km off in pinpointing the true locations of storms. A new generation of weather satellites will boost this accuracy by 2 to 4 times. The first in this new installment of NOAA's Geostationary Operational Environmental Satellites series, called GOES-N, was launched May 24 by NASA and Boeing for NOAA (National Oceanic and Atmospheric Administration). (A new polar-orbiting weather satellite, NOAA-18, was launched May 2005.)

Along with better accuracy at pinpointing storms, GOES-N sports a raft of improvements that will enhance our ability to monitor the weather—both normal, atmospheric weather and “space weather.”

“Satellites eventually wear out or get low on fuel, so we've got to launch new weather satellites every few years if we want to keep up the continuous eye on weather that NOAA has maintained for more than 30 years now,” says Thomas Wrublewski, liaison officer for NOAA at NASA's Goddard Space Flight Center. Currently, GOES-N is in a “parking” orbit at 90° west longitude over the equator. For the next 6 months it will remain there while NASA thoroughly tests all its systems. If all goes well, it will someday replace one of

the two active GOES satellites—either the eastern satellite (75°W) or the western one (135°W), depending on the condition of those satellites at the time. Unlike all previous GOES satellites, GOES-N carries star trackers aboard to precisely determine its orientation in space. Also for the first time, the storm-tracking instruments have been mounted to an “optical bench,” which is a very stable platform

that resists thermal warping. These two improvements will let scientists say with 2 to 4 times greater accuracy exactly where storms are located. Also, X-ray images of the Sun taken by GOES-N will be about twice as sharp as before. The new Solar X-ray Imager (SXI) will also automatically identify solar flares as they happen, instead of waiting for a scientist on the ground to analyze the images.

Flares affect space weather, triggering geomagnetic storms that can damage communications satellites and even knock out city power grids. The improved imaging and detection of solar flares by GOES-N will allow for earlier warnings. So for thunderstorms and solar storms alike, GOES-N will be an even sharper eye in the sky.

Find out more about GOES-N at goespoes.gsfc.nasa.gov/goes. Also, for young people, the Scijinks Weather Laboratory at scijinks.nasa.gov now includes a printable booklet titled “How Do You Make a Weather Satellite?” Just click on Technology.

This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.



New GOES-N satellite launches, carrying an imaging radiometer, an atmospheric sounder, and a collection of other space environment monitoring instruments.

An Invitation to SJAA Members

On Saturday, August 26, 2006, the Fremont Peak Observatory Association (FPOA) will be celebrating its 20th anniversary. Because the idea for the FPOA was born in the SJAA and the many cross memberships, I would like to extend a formal invitation to all SJAA members to attend the FPOA 20th Anniversary Party (Star-B-Que) at the observatory in Fremont Peak State Park.

For this occasion the FPOA plans to have the Star-B-Que catered. No final decision on the menu has been made, but the FPOA is considering barbecues and/or baked chicken. We shall also have specials for vegetarians. Each group or family should bring a dish to pass around. A larger than normal raffle with several grand prizes is planned. Also, as usual, we plan to have a well-known speaker for the event.

If you would like to attend, please make a reservation. You can logon and send an E-mail to schedule@fpoa.net or send an E-mail to me at kungfugina@aol.com. You can also leave a message on the observatory's voice mail at 831-623-2465. Please leave your name, number of people in your party, and that you are an SJAA member. If you are a member of FPOA, please note that also, so we don't count you twice. Finally, if you send an E-mail, please copy Mike Koop, so that he has an idea of the number of SJAA members that plan to attend.

Hope to see you there.

Patrick Donnelly, President FPOA

Solar System Stats for August 2006

Adapted from the Observer's Handbook published by The Royal Astronomical Society of Canada which in turn gets this data from the U.S. Naval Observatory's Nautical Almanac Office and Her Majesty's Nautical Almanac Office and contributions by David Lane, St. Mary's University, Halifax NS.

		Mercury	Venus	Mars	Jupiter	Saturn	Uranus	Neptune	Sun
RA	1	7 ^h 31 ^m	7 ^h 07 ^m	10 ^h 31 ^m	14 ^h 31 ^m	9 ^h 07 ^m	23 ^h 02 ^m	21 ^h 24 ^m	8 ^h 44 ^m
	11	8 ^h 07 ^m	7 ^h 59 ^m	10 ^h 55 ^m	14 ^h 34 ^m	9 ^h 12 ^m	23 ^h 01 ^m	21 ^h 23 ^m	9 ^h 22 ^m
	21	9 ^h 17 ^m	8 ^h 50 ^m	11 ^h 18 ^m	14 ^h 39 ^m	9 ^h 17 ^m	23 ^h 00 ^m	21 ^h 22 ^m	10 ^h 00 ^m
Dec	1	18°38'	22°25'	10°24'	-13°48'	17°25'	-7°03'	-15°27'	18°08'
	11	19°42'	20°56'	7°59'	-14°07'	17°03'	-7°11'	-15°32'	15°24'
	21	17°10'	18°27'	5°28'	-14°29'	16°41'	-7°20'	-15°37'	12°16'
Dist	1	0.75	1.54	2.50	5.28	10.16	19.25	29.05	1.015
	11	0.99	1.58	2.53	5.44	10.16	19.16	29.04	1.014
	21	1.23	1.61	2.56	5.59	10.14	19.11	29.06	1.012
Mag	1	1.3	-3.8	1.8	-2.1	0.3	5.8	7.8	
	11	-0.5	-3.8	1.8	-2.0	0.3	5.7	7.8	
	21	-1.3	-3.8	1.8	-2.0	0.4	5.7	7.8	
Size	1	9.0"	10.9"	3.7"	37.3"	16.4"	3.6"	2.3"	31'31"
	11	6.8"	10.6"	3.7"	36.2"	16.4"	3.7"	2.3"	31'34"
	21	5.5"	10.3"	3.7"	35.3"	16.4"	3.7"	2.3"	31'37"

Officers and Board of Directors

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Tres Gary Mitchell (408) 265-2336
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Publication Statement

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P.O. Box 28243
San Jose, CA 95159-8243

Submit

Submit articles for publication in the SJAA Ephemeris. Send articles to the editors via e-mail to ephemeris@sjaa.net. **Deadline, 10th of previous month.**

SJAA loaner scope status

All scopes are available to any SJAA member; contact Mike Koop by email (koopm@best.com) or by phone at work (408) 473-6315 or home (408) 446-0310 (Please leave message, phone screened).

Available scopes

These are scopes that are available for immediate loan, stored at other SJAA members homes. If you are interested in borrowing one of these scopes, please contact Mike Koop for a scope pick up at any of the listed SJAA events.

# Scope	Description	Stored by
1	4.5" Newt/ P Mount	Annette Reyes
3	4" Quantum S/C	Hsin I. Huang
6	8" Celestron S/C	Karthik Ramamurthy
7	12.5" Dobson	Tom Fredrickson
10	Star Spectroscope	Jim Albers
11	Orion XT6 Dob	Ravi Shankar Erram
12	Orion XT8 Dob	Sarah E. Jones
13	Orion XT6 Dob	Rajiv Vora
14	8" f/8.5 Dob	Colm McGinley
15	8" f/9 Dobson	Mike Koop
19	6" Newt/P Mount	Daryn Baker
23	6" Newt/P Mount	Wei Cheng
24	60mm Refractor	Al Kestler
26	11" Dobson	Vivek Kumar
27	13" Dobson	Steve Houlihan
29	C8, Astrophotography	Mark Ziebarth
32	6" f/7 Dobson	Sandy Mohan
34	Dynamax 8" S/C	Yuan-Tung Chin
35	Meade 8" Equatorial	Mike Horzewski
38	Meade 4.5" Digital Newt	Tej Kohli
39	17" Dobson	Steve Nelson
40	Super C8+	Bill Kerns
42	11x80 Binoculars	Ritesh Vishwakarma
43	Orion XT4.5 Dob	Gary Mitchell

Scope loans

These are scopes that have been recently loaned out. If you are interested in borrowing one of these scopes, you will be placed on the waiting list until the scope becomes available after the due date.

# Scope	Description	Borrower	Due Date
8	14" Dobson	Charles Santori	10/2/06
28	13" Dobson	Craig Scull	8/1/06
33	10" Deep Space Explorer	Art Kalb	9/30/06
36	Celestron 8" f/6 Skyhopper	Kristi Whitfield	9/16/06
41	18" Sky Designs Dob	Len Bradley	9/17/06

Extended scope loans

These are scopes that have had their loan period extended. If you are interested in borrowing one of these scopes, we will contact the current borrower and try to work out a reasonable transfer time for both parties.

# Scope	Description	Borrower	Due Date
2	6" f/9 Dob	John Paul De Silva	?
9	C-11 Compustar	Bill Maney	Indefinite
16	Solar Scope	Mike Koop	Repair
21	10" Dobson	Michael Dajewski	Repair
37	4" Fluorite Refractor	Peter Young	8/11/06
44	4.5" Skyview/ P Mount	Mantle Yu	8/03/06

Waiting list: Empty! Reserve a scope today!

San Jose Astronomical Association Membership Form

You can join or renew with the SJAA online at <http://www.sjaa.net/SJAAMembership.html>

☐ **New** ☐ **Renewal** (Name only, plus corrections below)

Membership Type:

- ☐ Regular — \$20
☐ Regular with Sky & Telescope — \$53
☐ Junior (under 18) — \$10
☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$10 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

Bring this form to any SJAA Meeting
or send (with your check) to:

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Make your check payable to "SJAA"
(not Sky Publishing).

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____

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